Advanced Hyperparameter Optimization

Chapter Summary

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Overview: Chapter 5 – Advanced Hyperparameter Optimization

This chapter covered the following topics:

1. **The Big Picture** – Why standard BO leaves efficiency on the table
2. **Simple Warmstarting** – Reusing prior observations from the same task
3. **Advanced Warmstarting** – Meta-learning portfolios across tasks
4. **Multi-Fidelity Optimization** – Cheap proxies to screen configurations early
5. **Priors in Bayesian Optimization** – Encoding expert knowledge into BO

Central theme: HPO is *not* a generic black-box problem –
exploit its structure (task similarity, cheap proxies, prior knowledge) to be more efficient.

1. The Big Picture

Standard BO is sequential, model-based, and theoretically grounded – but may still be too slow for expensive HPO tasks (e.g., large neural networks).

HPO has exploitable structure that generic black-box optimization ignores:

Observation	**Structure**	**Opportunity**
Same space, different datasets	Task similarity	Warmstarting from meta-data
Prior experiments available	Prior information	Informed initialization / priors
Training is iterative	Cheap approximations	Multi-fidelity screening

$\rightsquigarrow$ Each section in this chapter derives an HPO-specific extension of BO that exploits one of these structures.

2. Simple Warmstarting

Setting: Prior observations $\{(\lambda_i, c_i)\}$ are available from the *same* task.

How to use them:

- **Replace / augment the initial design $\mathcal{D}_{init}$** with existing observations: reduces initial uncertainty, biases the surrogate toward known good regions, and accelerates early exploitation
- **Tune surrogate hyperparameters:** use prior observations to set kernel length scales, estimate noise, or choose a better model family
- **Adjust the search space:** restrict Λ to regions supported by prior knowledge

Benefit	**Risk**
Faster convergence in early iterations	Misleads BO if prior is inconsistent or biased
Works without any meta-data machinery	Does not transfer across different tasks

3. Advanced Warmstarting: Portfolios Across Tasks

Setting: Meta-datasets $\mathbf{D}_{meta} = \{\mathcal{D}_1, \ldots, \mathcal{D}_{t-1}\}$ with prior HPO observations; new target task $\mathcal{D}_t$.

Goal: Build a portfolio $\mathbf{P} \subseteq \mathbf{\Lambda}_{\text{cand}}$, $|\mathbf{P}| = k$ that replaces the initial design for $\mathcal{D}_t$.

	Task-Agnostic	**Task-Specific**
Objective	$\min_{\mathbf{P}} \sum_i \min_{\boldsymbol{\lambda} \in \mathbf{P}} c(\boldsymbol{\lambda}, \mathcal{D}_i)$	Best configs on k-closest tasks
Construction	Greedy forward selection	Find similar tasks, pool their best configs
Similarity	Not needed	Meta-features, rankings, or learned embeddings
Build once?	Yes (offline)	No (per target task)
Weakness	May miss task-specific optima	Requires good meta-features; negative transfer if similarity is wrong

4. Multi-Fidelity Optimization

Idea: Approximate $c(\boldsymbol{\lambda})$ with a cheap proxy $c_b(\boldsymbol{\lambda})$, $b \in [b_{\min}, b_{\max}]$ (e.g., fewer data, fewer epochs, smaller model). Rank configurations early; allocate full budget only to survivors.

Successive Halving: Start with n random configs at budget $b_{\min}$; keep top $1/\eta$ fraction; multiply budget by η; repeat.

- η controls aggressiveness: large η = fast pruning; small η = gentle pruning
- Risk: good configs may be pruned due to noise at low fidelity

Hyperband: Run SH with multiple brackets (different n, $b_{\min}$ pairs) under equal total budget $\rightsquigarrow$ robust to the unknown optimal allocation.

BOHB (Falkner et al. 2018): Replace random sampling in SH/Hyperband with BO $\rightsquigarrow$ focuses search on promising regions while still pruning early.

Further extensions: Asynchronous Hyperband (no synchronization barrier), learning-curve predictors, scaling-law extrapolation.

4. Multi-Fidelity: Assumptions & When It Helps

Multi-fidelity methods rest on three assumptions:

1. **Error shrinks smoothly** as budget b increases
2. **Rankings are approximately preserved** across fidelities – a config that is bad at b_{min} is likely bad at b_{max}
3. **Higher fidelity gives a more accurate approximation** of the true objective

These assumptions can fail in practice – e.g., learning curves that cross, regularization effects that only appear at large budgets.

Method	**Adaptive sampling**	**Adaptive budget allocation**
Successive Halving	No (random)	Yes
Hyperband	No (random)	Yes (multiple brackets)
BOHB	Yes (BO)	Yes

5. Priors in Bayesian Optimization

Motivation: Domain knowledge should bias the search – standard BO ignores this.

Where priors enter BO:

- **Initialization:** seed the archive with informed configurations (Sections 2 & 3)
- **Surrogate model – implicit priors:**
  - *Kernel family*: encodes smoothness (RBF $\rightarrow$ very smooth; Matérn $\rightarrow$ rougher)
  - *Mean function* $\mu(\boldsymbol{\lambda})$: encodes where good values are expected
  - *Kernel hyperparameters* (length scales, noise): learned online; encode scale of variation
- **Acquisition function – explicit prior** (πBO, Hvarfner et al. 2022):

$$\boldsymbol{\lambda}_{n+1} = \arg\max_{\boldsymbol{\lambda} \in \boldsymbol{\Lambda}} \; u(\boldsymbol{\lambda}) \cdot \pi(\boldsymbol{\lambda})^{\beta/n}$$

Prior $\pi(\boldsymbol{\lambda})$ biases proposals; the β/n decay ensures exploration recovers as n grows

Extensions: data-driven priors from past HPO runs; model-driven priors from LLMs or pre-trained foundation models (e.g., Optformer).

Chapter Summary

Key takeaways from Chapter 5:

1. Standard BO is powerful but treats HPO as a generic black-box
2. **Simple warmstarting** reuses prior observations from the same task to seed the initial design and tune the surrogate; biased priors can mislead
3. **Portfolio-based warmstarting** transfers knowledge across tasks
4. **Multi-fidelity methods** screen many configurations cheaply and invest full budget only in survivors: SH, Hyperband, and BOHB rely on rank preservation across fidelities
5. **Priors** steer BO implicitly (kernel choice, mean function, length scales) or explicitly (πBO); prior mismatch hurts efficiency

$\rightsquigarrow$ The more structure we exploit, the more efficient HPO becomes – but every assumption introduces a new failure mode.